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Key Points:

- The structure and magnitude of vertical mixing varied dramatically over the annual stratification cycle in a large, deep lake
- Density stratification limited mixing over the summer, while weak velocity shear restricted mixed layer growth during the transitional fall
- Radiative convection and wind shear drove full water-column mixing over the convective spring, which occurred locally for >100 days/year

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Characterizing the Seasonal Variability of Hypolimnetic Mixing in a Large, Deep Lake

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Abstract In this study, we report on turbulent mixing observed during the annual stratification cycle in the hypolimnetic waters of Lake Michigan (USA), highlighting stratified, convective, and transitional mixing periods. Measurements were collected using a combination of moored instruments and microstructure profiles. Observations during the stratified summer showed a shallow, wind-driven surface mixed layer (SML) with locally elevated dissipation rates in the thermocline ($\epsilon \approx 10^{-7}$ W / kg) potentially associated with internal wave shear. Below the thermocline, turbulence was weak ($\epsilon \approx 10^{-9}$ W / kg) and buoyancy-suppressed ($Q = \epsilon / \nu N^2 < 8.5$), with low hypolimnetic mixing rates ($K_z \approx 10^{-7}$ m^2 / s) limiting benthic particle delivery. During the convective winter period, a diurnal cycle of radiative convection was observed over each day of measurement, where temperature overturns were directly correlated with elevated turbulence levels throughout the water column ($\epsilon \approx 10^{-7}$ W / kg; $K_z \approx 10^{-1}$ m^2 / s). A transitional mixing period was observed for spring conditions when surface temperatures were near the temperature of maximum density (T_{MD}≈3.98°C) and the water column began to stably stratify. While small temperature gradients allowed strong mixing over the transitional period ($K_Z \approx 10^{-3}$ m^2 / s), hypolimnetic velocity shear was overwhelmed by weakly stable stratification ($N^2 \approx 10^{-6}$ s^{-2}; $Ri \approx 0.2$), limiting the development of the SML. These results highlight the importance of radiative convection for breaking down weak hypolimnetic stratification and driving energetic, full water column mixing during a substantial portion of the year (>100 days at our sample site). Ongoing surface water warming in the Laurentian Great Lakes is significantly reducing the annual impact of convective mixing, with important consequences for nutrient cycling, primary production, and benthic-pelagic coupling.

Plain Language Summary Turbulent mixing is a fundamental physical process in aquatic systems, where it plays a role in transferring momentum and heat, regulating water quality, and controlling the vertical transport of nutrients and dissolved gases. Mixing rates in large, temperate lakes are especially dynamic, since annual changes in atmospheric forcing (e.g., air temperature, wind speed, daylight hours, etc.) have dramatic consequences for both the magnitude and structure of turbulence throughout the water column. In this study, we investigate annual changes in deep water mixing as observed at a 55 m deep site in a large, freshwater lake (Lake Michigan, USA). Observations are used to define three distinct mixing regimes, including (a) the stratified period, where strong density gradients suppressed turbulent overturns; (b) the transitional period, where the water column was nearly isothermal and wind-driven mixing was only marginally affected by density stratification; and (c) the convective period, where radiative heating generated density instabilities that enhanced mixing across the water depth. Results are used to describe the effects of recent surface warming on reducing deep-water mixing. This work represents one of the most comprehensive studies of offshore mixing in the Laurentian Great Lakes, with important implications for biophysical modeling and invasive species management.

1. Introduction

In temperate lakes, surface forcing and density stratification vary dramatically throughout the year, creating seasonal differences in turbulent mixing characteristics for both nearshore and offshore waters (Wüest & Lorke, 2003). For large, dimictic freshwater lakes such as the Laurentian Great Lakes, this seasonality falls into three broad categories: (a) the stratified summer; (b) the (*weakly*) inversely stratified winter, when surface temperatures are below the temperature of maximum density (T_{MD} = 3.98 °C for freshwater) and partial or complete ice cover may be present; and (c) an approximately isothermal spring and fall period during which

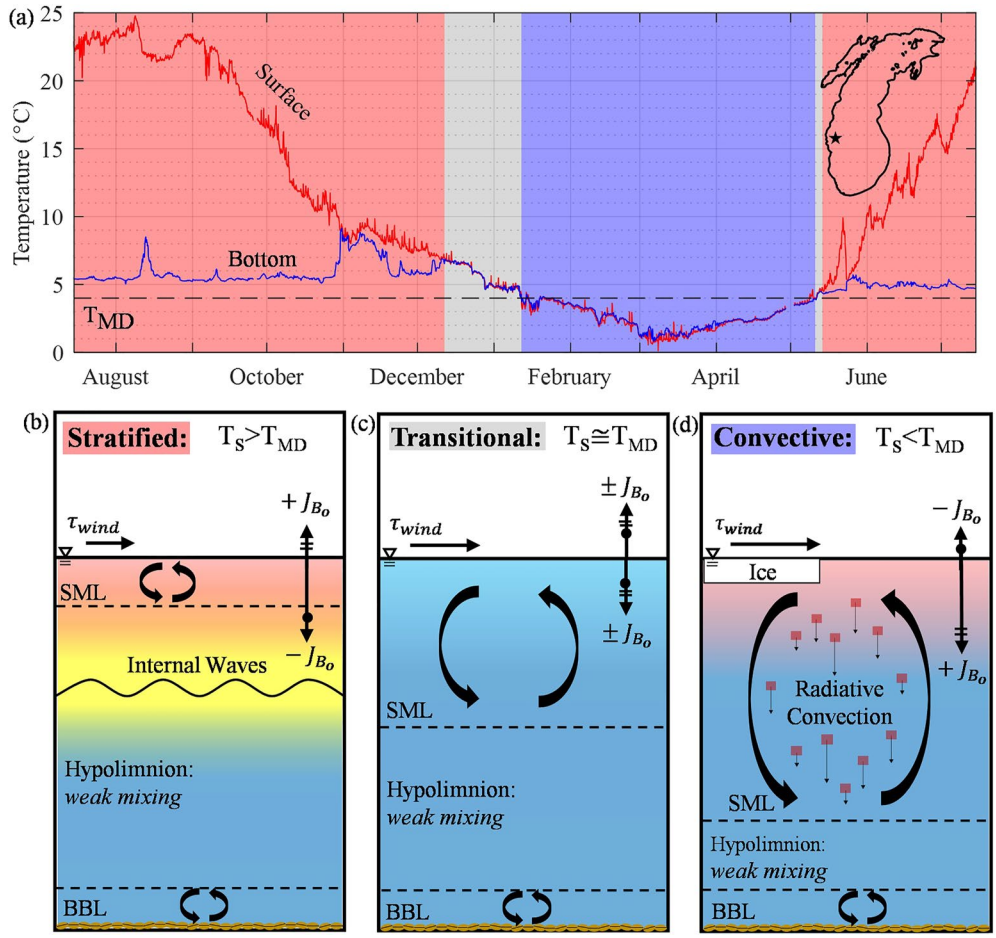


Figure 1. Annual cycle of thermal stratification and mixing in deep, dimictic lakes, as illustrated by surface (T_s) and bottom temperature measurements from 2012 to 2013 (Lake Michigan: 55 m depth; Cannon & Troy, 2018) and water column schematics for the stratified (b), transitional (c), and convective (d) periods. The inset map shows Lake Michigan and the location (star) where measurements were collected (both in this study and in Cannon & Troy, 2018) Background colors in (a) represent the mixing regimes described in (b)–(d), where red, gray, and blue show the stratified, transitional, and convective periods, respectively. The temperature of maximum density (T_{MD} = 3.98 °C for pure freshwater) is indicated with a dashed horizontal line. Regions of strong mixing are shown with circular arrows, and arrow lengths are used to indicate the relative magnitude of each external forcing mechanism (wind shear: τ_{wind}; surface flux: J_{B_o}). Stabilizing (destabilizing) surface fluxes are indicated with dotted (double-dashed) arrows and negative (positive) flux notation.

the lake transitions between stratified and inversely stratified. This seasonal cycle is demonstrated in Figure 1, which shows year-round mooring data from our 55 m depth site in Lake Michigan, USA, the focus of this study.

While near-surface mixing is largely driven by wind stress and surface convection during all three periods (Imberger & Hamblin, 1982), deep hypolimnetic mixing is complicated by additional physical processes, especially density stratification. The cyclical effects of turbulence-suppressing density stratification (both summer and winter) are impacted by unique seasonal generation mechanisms, including internal waves (Bouffard et al., 2012) and deep radiative convection (Austin, 2019; Cannon et al., 2019). Understanding and parameterizing this annual cycle of hypolimnetic mixing is imperative for developing accurate biogeochemical models in large lakes, for which seasonal differences in mixing rates have significant consequences for particulate transport (Eadie & Robbins, 1987), primary production (Fahnenstiel et al., 2010), and deep-water oxygen renewal (Piccolroaz & Toffolon, 2013).

In the Laurentian Great Lakes, the annual cycle of hypolimnetic mixing is especially important for its role in benthic-pelagic nutrient exchange and food availability for invasive dreissenids (e.g., quagga mussels). Since

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Table 1
Summary of All Experiments Used to Characterize Hypolimnetic Mixing in the Current Study

Name	Description	Mooring dates	Microstructure dates
Stratified	Stably stratified with surface temperatures above T_{MD}.	August 1–16 [2017]	August 1, 2, 9, 10, 15, and 16
Transitional	Nearly isothermal; surface temperatures approximately equal to T_{MD}; $T_S - T_B < 1$°C	May 13 to June 4 [2019]	May 13 and 14
Convective	Inverse stable stratification at night coupled with isothermal daily conditions; no ice cover; $0 < T_S < T_{MD}$	April 5–20 [2018]	April 5, 12, and 20

Note. The instrumented tripod was deployed over all indicated mooring dates, while microstructure measurements were only collected periodically. T_S is the surface temperature, T_B is bottom temperature, and T_{MD} is the temperature of maximum density, taken as 3.98°Cat the surface in pure freshwater.

their initial introduction in the early 1990s, invasive quagga mussels (*Dreissena rostriformis bugensis*) have been linked to significant ecosystem changes in the lower Great Lakes, including native species displacement (e.g., *Diporeia*; Nalepa et al., 2009), increased water clarity (Barbiero et al., 2018), and decreases in primary production and spring phytoplankton biomass (Fahnenstiel et al., 2010; Vanderploeg et al., 2010). While quagga mussels are prolific filter feeders, with filtration rates as high as 5.4 L/day/mussel (Mosley & Bootsma, 2015), they rely entirely on the hydrodynamic delivery of particulates for survival (Ackerman et al., 2001), with ecosystem impacts that are directly set by the structure and magnitude of hypolimnetic turbulence (e.g., Koseff et al., 1993). Although recent mussel *modeling* efforts have highlighted the importance of seasonal variations in hypolimnetic mixing (e.g., Rowe et al., 2015), supporting turbulence observations in deeper waters remain scarce, with measurements generally restricted to shallow stratified systems (e.g., Boegman et al., 2008; Ivey & Patterson, 1984; Loewen et al., 2007; Rao et al., 2014) and benthic boundary layers (Cannon & Troy, 2018). Quagga mussel populations continue to expand in the deeper waters of the Laurentian Great Lakes (Glyshaw et al., 2015) and are threatening deep lakes and reservoirs in the southwestern United States and Europe (Heiler et al., 2013; Lindsay et al., 2018). The development of useful biophysical models for these deep systems (e.g., Shen et al., 2018) requires a solid understanding of, and an ability to parameterize, the annual cycle of hypolimnetic mixing in large, deep lakes.

In this study, we utilize direct measurements of turbulent mixing to describe the seasonal variability of hypolimnetic mixing at a deep water (55 m) site in Lake Michigan for which we have previously reported on bottom boundary layer (Cannon & Troy, 2018) and convective processes (Cannon et al., 2019). A combination of high-resolution acoustic Doppler velocimetry and periodic microstructure profiling was used to quantify mean flow and turbulence throughout the water column, with deployments designed to resolve both the vertical and temporal variability of hypolimnetic mixing in the lake. Observations are used to describe mixing during the stratified summer, convective winter, and transitional spring, providing insight into mixing characteristics over much of the annual stratification cycle, with the exception of the transitional fall when water conditions were not safe for data collection. Importantly, this research presents some of the first *direct* measurements of hypolimnetic mixing in Lake Michigan, where previous studies have focused almost exclusively on hydrodynamic modeling validated by temperature and velocity profile measurements (e.g., Rowe et al., 2015; Shen et al., 2018). The real-world measurements presented herein provide a valuable benchmark for both hydrodynamic and biophysical modeling in large temperate lakes.

2. Methods

2.1. Observations

Measurements of water velocities, thermal stratification, and turbulent microstructure were collected at a 55 m depth site in Lake Michigan near Milwaukee, WI, USA (43° 04′27′′N, 87° 45′14′′W) between August 2017 and July 2019. Observations consisted of three separate deployments (Table 1), each approximately 2 weeks long, and were designed to resolve seasonal variability associated with different stratification and surface forcing conditions. All observations were divided into three unique mixing regimes, defined as the stratified, transitional, and convective periods (Table 1). The 55-m-deep measurement site was located on the outer edge of the lateral coastal boundary layer (Rao & Schwab, 2007) and was selected such that a significant portion of the water column would be unaffected by boundary layer stress. Additionally, at a distance

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of 9 km offshore, it is beyond the internal Rossby radius (typically 3–5 km; Beletsky et al., 1997) and thus unaffected by internal Kelvin waves during the stratified period, with the exception of an occasional large-scale upwelling event. We have previously reported both physical and biogeochemical characteristics for this site (Cannon & Troy, 2018; Cannon et al., 2019; Mosley & Bootsma, 2015; Shen et al., 2018; Troy et al., 2016).

Mean currents and velocity shear were measured using a combination of three high-resolution acoustic Doppler velocimeters. Mean velocities in the bottom boundary layer (10–140 cm above bottom) were measured using a Nortek Aquadopp HR profiler (sample rate: 2 Hz; bin size: 2–3 cm) and a Nortek Vector (sample rate: 16 Hz; sample volume: 1.5 cm), while full water column velocities and mean shear (~2–45 mab) were estimated using a five-beam Nortek Signature 500 (sample rate: 2–4 Hz; bin size: 1–2 m). For the stratified and convective experiments, all three instruments were deployed concurrently on a large (~2 m tall) instrumented tripod to measure conditions continuously over several weeks. The Signature 500 was deployed alone on a small frame during the transitional experiment, with the nearest-bed velocity measurement occurring at 1.5 mab. The low signal-to-noise ratio from the Signature 500 data for all deployments precluded its use for turbulent statistics but the data quality was sufficient for computing reliable "mean" (10 min average) current speeds, mean shear and Richardson number profiles, and near-bed logarithmic/log-linear velocity profile fitting during high-speed episodes.

Water column temperatures were measured using a high-resolution temperature chain, equipped with 12 RBR TR-1060 thermistors (accuracy: ±0.002°C; sample rate: 1 Hz) and six SBE-56 thermistors (accuracy: ±0.002°C; sample rate: 2 Hz). The sensor spacing varied between deployments, but they were generally spaced such that the vertical resolution was maximized in regions where strong temperature gradients were expected (2 m spacing near thermocline; 1 m spacing within 5 m of boundaries) and the distance between individual thermistors was never larger than 4 m. Temperature chains were deployed for extended periods during the stratified and convective experiments, with water temperature measurements limited to the microstructure casts (discussed below) during the transitional period.

Microstructure profiling was performed periodically during each deployment using an RSI MicroCTD (Rockland Scientific International, Inc.; hereafter: "MCTD"). The MCTD measured the vertical shear of velocity (2× air-foil shear probe), micro-temperature (2× FP07 temperature probe), and pressure at 512 Hz, with additional conductivity (JAC-CT), fluorescence, and turbidity (JAC-OBS/FL) measurements recorded at 64 Hz. On-board vibration (two-axis) and tilt sensors were used for quality control and to correct the data for non-turbulent probe fluctuations. The instrument was profiled at 0.80 m/s resulting in an average vertical resolution of 1.5 mm. Although microstructure profiling was limited to periods of relatively weak wind forcing ($U_{10} < 10$ m/s), these conditions are not atypical. Multi-decadal (1979–2009) hindcast model data on Lake Michigan (USACE Wave Information Studies: http://wis.usace.army.mil/; Last Accessed: May 2019) show that lake-wide wind speed averages are under 10 m/s for the majority (>90%) of the year, suggesting that our measurements are more or less representative of typical forcing conditions. Details on microstructure processing are provided in a subsequent section.

The buoyancy frequency (N^2) was calculated from temperature and pressure measurements using estimated density profiles. In-situ water densities (ρ) were calculated from all microstructure and temperature chain measurements using the pressure-corrected freshwater equation of state described by Tanaka et al. (2001), which requires only temperature (T) and pressure (P) as input parameters. Periodically collected CTD profiles showed salinity was generally low (<0.2 ppt) and unimportant for water density estimates, as previously reported by Chapra et al. (2012). Density profiles were converted to local buoyancy frequencies following Gill (2016):

$$N^2 = -\frac{g}{\rho_o}\left[\frac{d\rho}{dz} + \frac{\rho_o g}{c_s^2}\right] \quad (1)$$

where $g = 9.81\,\mathrm{m/s^2}$ is the gravitational constant, ρ_o is a reference density (taken as the mean density between points of interest), and $c_s(T,P)$ is the local speed of sound (calculated with TEOS-10; McDougall & Barker, 2011). The reference pressure and speed of sound were calculated as the averages between adjacent sensors (or microstructure sub-samples). For microstructure profiles, the density gradient $\frac{d\rho}{dz}$ was estimated using a best-fit slope through the pressure (depth) and density measurements of each subsample, while

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a simple finite difference approach was used to estimate the density gradient between adjacent thermistors. Although this parametrization is slightly more complicated than typical potential density-based (ρ_θ) parameterizations ($N^2_{\rho_\theta} = -\frac{g}{\rho_\theta}\frac{d\rho_\theta}{dz}$), it has the advantage of more accurately estimating the strength of stability under weakly stable conditions (King et al., 2012), where water column stability can be dominated by pressure effects over relatively shallow depths (<50 m). The depth dependence of the temperature of maximum density (T_{MD}) was modeled following Bouffard and Wüest (2019), and T_{MD} decreased linearly from $T_{\mathrm{MD}} = 3.98$°C at the surface to $T_{\mathrm{MD}} = 3.88$°C at the lake bed (55 m depth).

For the stratified period, the surface mixed layer (SML) depth for individual microstructure profiles was estimated using a surface temperature difference threshold (e.g., MacKinnon & Gregg, 2005), with SML defined as the first depth at which water temperatures were more than 0.25 °C cooler than surface temperatures. For the transitional and convective periods, where temperature gradients were very small, the SML was defined as the first depth at which the turbulent kinetic energy dissipation rate (ϵ) was less than the background dissipation rate ($10^{-9}\ W\ /\ kg$; see discussion). This definition for SML is comparable to definitions of the *actively* mixed layer depth used in other lake (Tedford et al., 2014) and ocean (Sutherland et al., 2014) studies.

Atmospheric forcing characteristics (e.g., wind speed, wave height, air temperature, etc.) near the mooring were obtained from the Great Lakes Coastal Forecasting System (GLCFS: https://www.glerl.noaa.gov/res/glcfs/; Last Accessed: September 1, 2019), which uses interpolated buoy and land-based wind measurement stations to drive the GLERL-Donelan wave model (Schwab et al., 1984). These model parameters were validated on-site using an on-ship anemometer and nearby meteorological buoy (NDBC station 45013). Wind stresses (τ_w) and surface friction velocities ($u_* = \sqrt{\tau_w\ /\ \rho_{water}}$) were estimated using standard wind-water drag coefficients and estimated wind speeds at the mooring location (Wüest & Lorke, 2003), while incoming shortwave solar radiation (SW_o) was sourced from the North American Land Data Assimilation System (NLDAS: https://giovanni.gsfc.nasa.gov/giovanni/; Last Accessed: September 1, 2019) and compared to several land-based weather stations and limited on-site measurements for validation.

The effective mixed layer heat flux (H_*) was calculated over each deployment using parameterized longwave (LW), latent (H), and sensible (E) heat fluxes following the methods outlined in Tedford et al. (2014), such that:

$$H_* = H + E + LW_{IN} - LW_{OUT} + SW\left(SML\right) \tag{2}$$

Estimates of E and H were calculated following Verburg and Antenucci (2010), while incoming (LW_{IN}) and outgoing (LW_{OUT}) longwave radiations were estimated from atmospheric forcing parameters using the equations presented in Rubin and Atkinson (2001). Only the fraction of penetrative shortwave radiation absorbed within the SML is considered in H_*, with surface radiation SW_o attenuated over the depth using a two-layer exponential model, as discussed in Cannon and Troy (2018). For this study, best fit light extinction coefficients ($SW\left(z\right) = SW_o\left[0.36e^{-z/L_1} + 0.47e^{-z/L_2}\right]; L_1 = 0.71\,\mathrm{m};\ L_2 = 7.7\,\mathrm{m}$) showed that over 90% of the incoming shortwave radiation was absorbed within 12 m of the surface. The surface buoyancy flux was estimated from the effective mixed layer heat flux as $J_{B_o} = g\alpha H_*\ /\ c_p\rho$, where α is the thermal expansion coefficient and c_p is the specific heat of water. Using this definition, negative (positive) values represent stabilizing (destabilizing) surface buoyancy fluxes. Finally, the Monin-Obukhov length ($\mathrm{L_{MO}}$) was calculated using surface friction velocities and mixed layer heat fluxes, such that $L_{MO} = -u_*^3\ /\ \kappa J_{B_o}$ (Tedford et al., 2014).

2.2. Microstructure Analysis

The majority of the data analysis in this document is concerned with hypolimnetic turbulence parameters calculated using vertical microstructure profiles. For this study, 336 microstructure profiles were collected over 11 days using an RSI MicroCTD equipped with both microstructure shear (2× SPM-B-6000) and temperature (2× FP07) probes. The instrument was deployed by hand, free-falling at 80 cm/s, and retrieved using an onboard winch. A combination of shear and temperature microstructure was used to obtain estimates of both turbulent kinetic energy (ϵ) and turbulent scalar diffusivity (K_z) throughout the water column.

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Vertical microstructure profiles of temperature and velocity shear were sub-sampled at regular time intervals to create discrete time series for turbulence analysis. Each profile was broken into 2 s (1,024 samples) time increments overlapped by 50% (1 s), resulting in approximately 1.6 m bin sizes with a 0.8 m vertical resolution. Vehicle kinematics were used to filter the data, with all sub-samples showing excessive instrument tilt ($> \pm 5°$) or vibrations (two-axis) rejected from further analysis. Similarly, all segments with fall velocities less than a threshold value of 75 cm/s were discarded to remove accelerating and decelerating regions at the beginning and end of each profile.

Turbulent kinetic energy dissipation was estimated from velocity shear measurements using a combination of Nasmyth empirical spectrum fitting (Bluteau et al., 2016) and power spectral density function integration (Osborn, 1974), as described in Lueck (2015). The raw velocity shear for each segment (2 s; 1024 points) was parabolic-detrended and used to estimate energy spectra using 50% overlapped 0.5 s (256 points) windows and a cosine windowing function. Algorithms for calculating dissipation were provided by the manufacturer (Rockland Scientific, Inc.), and additional details for this method can be found in Lueck (2015). Following suggestions in Bluteau et al. (2016), the mean absolute deviation (MAD) was used to identify and discard all observed spectra that deviated significantly from shape of the model spectrum (over all n wavenumbers in the fit range). We used the oft-cited misfit criteria of $MAD > 2\left(2 / d\right)^{1/2}$, where d is the degrees of freedom of the observed spectra, to reject poor spectral fits (Ruddick et al., 2000). Finally, dissipation estimates were used to calculate the turbulence intensity, or isotropy, parameter $Q = \epsilon / \nu N^2$, with the kinematic viscosity ν estimated based on the local water temperature. Water column mixing was defined as molecular ($Q < 1.7$), buoyancy-controlled ($1.7 < Q < 8.5$), transitional ($8.5 < Q < 100$), or energetic ($Q > 100$) based on thresholds presented in Bouffard and Boegman (2013).

For the stratified and transitional periods, the turbulent thermal diffusivity (K_z), also known as the vertical scalar diffusivity, was estimated from temperature gradient spectrum using the best-fit temperature variance (χ) derived from empirical Batchelor function fitting. This method is based on the assumption that the water column is stably stratified, with turbulence driven by shear instabilities rather than convective mixing, an assumption supported by stabilizing surface fluxes observed during both sets of experiments. Raw temperature and pressure data were used to calculate the vertical gradient of temperature (dT / dz) for each microstructure profile. Each segmented series of temperature gradient data (2 s; 1,024 points) was linearly detrended and used to estimate energy spectra ($\Phi_{dT/dz}$) using 50% overlapped 0.5 s (256 points) Hamming windows (i.e., Welch's method). After instrument accelerations were removed (see above), a transfer function was used to correct all temperature spectra for the thermistor's frequency response (Bluteau et al., 2017). The corrected energy spectra of temperature gradient fluctuations were used to estimate the temperature variance (χ) by iteratively fitting the theoretical Batchelor spectrum to the non-noise dominated range of observed energy spectra using the maximum likelihood estimator approach described in Steinbuck et al. (2009). The Batchelor wave number ($k_B = \left(\epsilon / \nu D_T^2\right)^{1/4}$; see Ruddick et al., 2000) for each fit was computed using shear-derived dissipation estimates to reduce the number of fit parameters and eliminate the high fit uncertainty associated with Batchelor-fit dissipation (Jonas et al., 2003). As with the Nasmyth-Osborn approach for shear spectrum fitting, all fit spectra with $MAD > 2\left(2 / d\right)^{1/2}$ were removed from further analysis. Similarly, all other misfit criteria presented in Ruddick et al. (2000), including the log-likelihood ratio (LHR < 2) and signal-to-noise ratio (SNR < 1.3), were used to further improve the quality of temperature variance (χ) estimates. Turbulent scalar diffusivity was estimated directly from the dissipation rate of temperature variance (e.g., Bluteau et al., 2016), such that $K_z = \frac{\chi}{2}\left(\frac{dT}{dz}\right)^{-2}$. The mean temperature gradient dT / dz was estimated using a best-fit slope through all microstructure temperature and depth measurements in a single segment. Diffusivity estimates were further quality controlled by removing all microstructure segments with mean temperature gradients below the measurement threshold ($dT / dz < 2T_A / \Delta z$, where $T_A = 0.002$ °C is the sensor accuracy and Δz is the change in depth over the segment).

Unlike the stratified and transitional periods, mixing observed during the convective period was heavily influenced by radiative convection, as previously described in Cannon et al. (2019). The Batchelor method, which is derived assuming steady-state mixing and negligible vertical advection (Osborn & Cox, 1972), cannot be used to estimate turbulent scalar diffusivity under convective conditions. Instead, diffusivity was estimated "directly" using the buoyancy flux, such that:

$$K_z(z) = -\frac{B(z)}{g\alpha(z)\left[dT\ /\ dz(z)\right]} \tag{3}$$

where g is the gravitation constant, $\alpha = -\rho^{-1}\partial\rho\ /\ \partial T$ is the thermal expansion coefficient, $dT\ /\ dz$ is the vertical gradient of temperature, and $B(z)$ is the buoyancy flux. In this study, the buoyancy flux was estimated from temperature measurements using the 1-D heat budget, described in detail in Cannon et al. (2019). This method assumes that the observed temperature variations are driven entirely by vertical turbulent mixing and solar radiation, and neglects any heat exchanged with the lake bed (sediments). Diffusivity estimates from the 1-D heat budget method were restricted to periods of weak lateral advection, a condition that was met on all 3 days of microstructure profiling during the convective experiment.

Microstructure measurements within 3–5 m of the surface and bottom boundaries were often missing or removed due to vehicle kinematics issues (i.e., strong accelerations/decelerations). Turbulence estimates (i.e., ϵ, K_z) in these regions were instead parameterized using similarity scaling arguments. Turbulent kinetic energy dissipation was parameterized using the law-of-the-wall ($\epsilon_{LOW} = u_*^3\ /\ \kappa z$) at both boundaries, with near-surface friction velocities (u_*) estimated using the wind stress ($u_* = \sqrt{\tau_w\ /\ \rho_{air}}$) and near-bed friction velocities estimated from log-linear velocity profile fitting (e.g., Cannon & Troy, 2018). Near-bed turbulent scalar diffusivities were also parameterized using law-of-the-wall estimates ($K_{z-LOW} = u_*\kappa z$), while near-surface diffusivities were estimated using the model of Troen and Mahrt (1986), which was initially developed for use in atmospheric boundary layers and formed the basis of the popular KPP model developed by Large et al. (1994). Model diffusivities (K_{z-TM}) were estimating following:

$$K_{z-TM} = u_*\kappa z\phi_s^{-1}\left(1 - \frac{z}{SML}\right)^2 \tag{4}$$

where ϕ_s is the non-dimensional shear defined for stable ($L_{\mathrm{MO}} > 0$: $\phi_s = 1 + 5z\ /\ L_{MO}$) and unstable ($L_{\mathrm{MO}} < 0$ $\phi_s = \left(1 - 16z\ /\ L_{MO}\right)^{-1/2}$) surface buoyancy fluxes. This parameterization is limited to depths much smaller than the SML ($z \ll$ SML), where turbulence generation is expected to be dominated by wind stress and surface buoyancy fluxes.

Turbulence characteristic distributions were typically lognormal and required transformations to provide reasonable, unbiased estimates of the mean mixing conditions observed on each day of profiling. As such, the averages and 95% confidence intervals (CIs) for turbulence variables (ϵ, K_z, Q, etc.) presented in this study were computed using the arithmetic mean of log-transformed measurements. Additionally, all CIs were estimated using bootstrapping to avoid errors associated with distribution assumptions.

3. Results

Observations from each mixing period are presented below. Background conditions (wind, currents, and temperatures) for each period are shown in Figure 2, with representative vertical profiles of background flow, stratification, and estimated turbulent quantities collected on a single day of microstructure profiling shown in Figure 3. Figure 4 shows the daily averaged turbulent profiles to illustrate the variability observed within the three periods. Observations are discussed separately for each period (stratified, transitional, and convective) and then synthesized in the discussion section.

3.1. Stratified Period

The stratified period was characterized by stable density stratification across the entire water column, with warm surface waters ($T_{\mathrm{S}} \approx 20$°C) separated from cooler hypolimnetic waters ($T_{\mathrm{B}} \approx 5$°C) by a persistent thermocline ($N^2_{max} \approx 10^{-3}\mathrm{rad}^2\ /\ \mathrm{s}^2$) that fluctuated between 10 and 20 m depth (Figure 2g). Near-surface currents rarely exceeded 20 cm/s, with maximum values (5 m depth: 33 cm/s) corresponding to periods of elevated wind stresses. While hypolimnetic mean velocity shear was highest during the stratified period ($S^2 = 10^{-5} - 10^{-4}\mathrm{s}^2$), strong stratification led to Richardson number ($Ri = N^2/S^2$) estimates that exceeded 1/4 throughout the water column, with the exception of the bottom 10–15 m where Ri<1/4 (Figure 3c). Horizontal velocities and thermocline fluctuations were dominated by near-inertial energy (period = 17.5 h;

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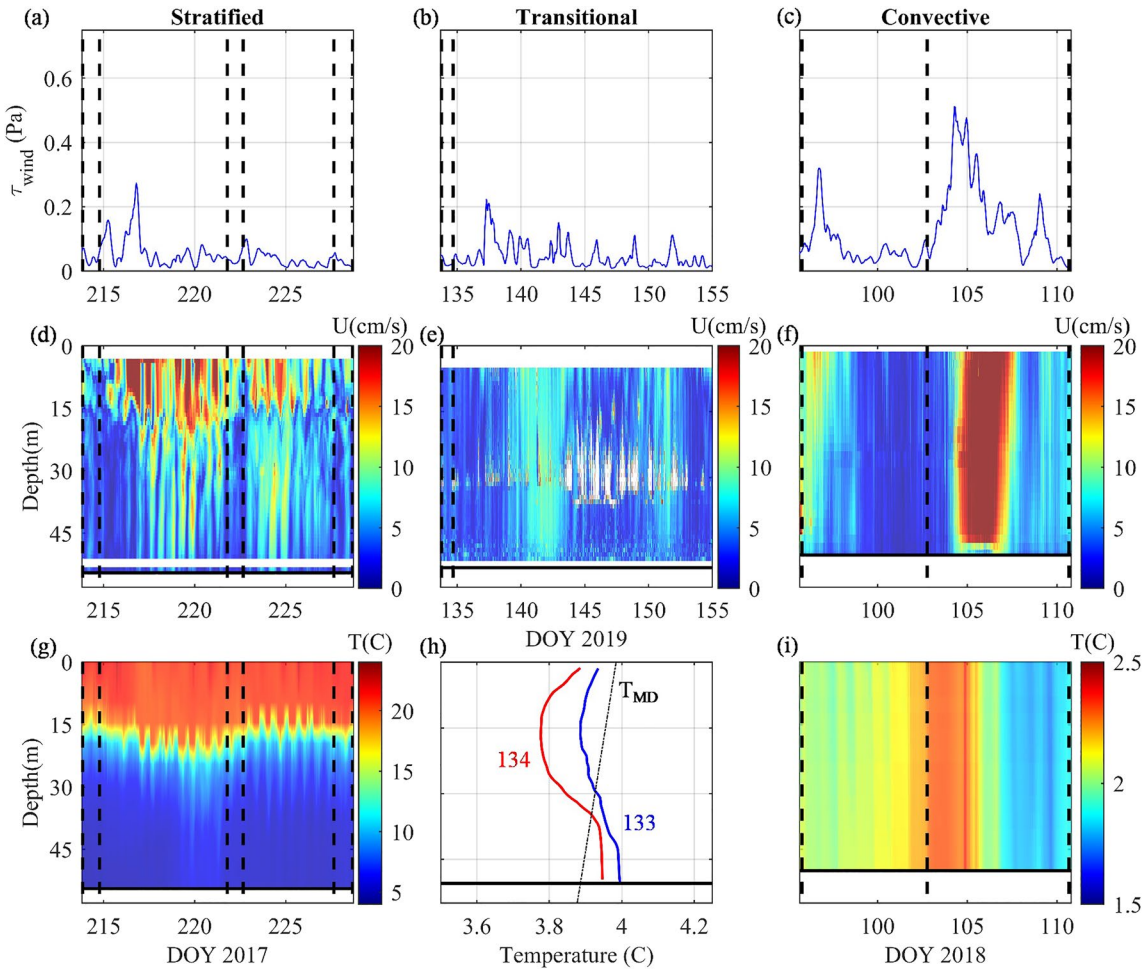


Figure 2. Water column overview including conditions as observed over the stratified (a, d, g), transitional (b, e, h), and convective (c, f, i) periods. Subplots include wind shear (a)–(c), current speeds (d)–(f), and water column temperatures (g)–(i). Temperature observations were limited to microstructure casts during the transitional period (h; blue: DOY 134; red: DOY 133). Periodic microstructure measurements are indicated with vertical dashed lines and the variable sample site depth is shown with a horizontal line on each subplot. The black line in (h) shows the temperature of maximum density (T_{MD}) as a function of depth.

Figure 5), with velocity signals above and below the thermocline that were 180° out of phase. This agreed well with the analysis of Choi et al. (2012), who showed that near-inertial internal Poincaré waves dominate the variability of currents and temperatures in the offshore waters of Lake Michigan throughout the stratified summer, with a maximum influence during August (i.e., during this deployment period).

Microstructure measurements showed that density stratification was sufficiently strong to suppress turbulence outside the surface and bottom boundary layers during the stratified period (Figures 3a–3f). Mixing was most energetic near the surface and bottom boundaries, where turbulent intensity parameters exceeded the threshold for energetic mixing ($Q > 100$). SML depths ranged from 4 to 10 m, with the most energetic mixing conditions linked to strong wind forcing (DOY 222: $\tau_w = 0.12$ Pa; $K_{z-5m} = 6 \times 10^{-3} \mathrm{m}^2$ / s). Near the bed, observed dissipation rates and diffusivities were consistently elevated above hypolimnetic levels within 5–7 m of the bottom, the same region over which $Ri <$ ¼. These benthic turbulence levels were comparable to our own previous velocimeter-based turbulence estimates at the same site (Cannon & Troy, 2018; Troy et al., 2016), although turbulence estimates were limited to less than 1m above the bed for those studies. Turbulent intensities were within the buoyancy controlled ($1.7 < Q < 8.5$: thermocline) or transitional ($8.5 < Q < 100$: hypolimnion) regimes for all measurements between 10 and 40 m depth. Below the thermocline and above the bottom boundary layer, hypolimnetic dissipation was nearly constant at 10^{-9} W / kg, with similarly weak vertical mixing ($\overline{K_z} = 3 \times 10^{-6} \mathrm{m}^2$ / s). Although dissipation profiles were

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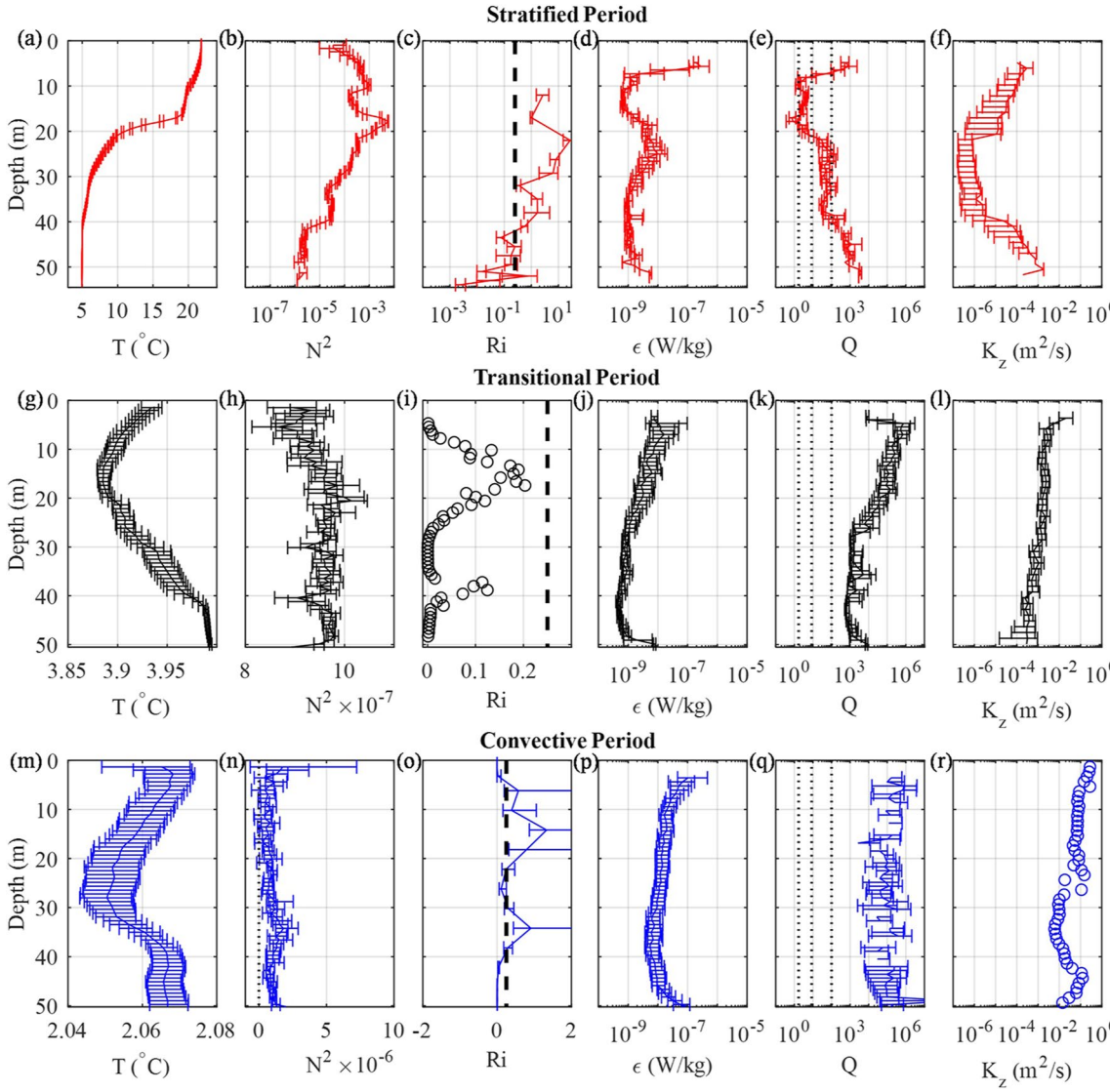


Figure 3. Example daily averages and 95% confidence intervals for turbulent mixing parameters observed during each deployment period (stratified: (a–f); transitional: (g–l); convective: (m–r)), including (a, g, m) temperature; (b, h, n) buoyancy frequency N^2; (c, i, o) gradient Richardson number; (d, j, p) turbulent kinetic energy dissipation ϵ; (e, k, q) turbulent intensity parameter $Q = \epsilon / \nu N^2$; and (f, l, r) turbulent scalar diffusivity K_z. Vertical reference lines are included for the critical Richardson number ($Ri = 1/4$) and the turbulence regimes (Q) defined in Bouffard and Boegman (2013).

often characterized by local maxima in the thermocline due to baroclinic shear ($\epsilon \approx 10^{-8}$ W / kg), estimated thermocline diffusivities were small due to the offsetting effect of strong density stratification. Mixing efficiencies ($\eta = B / \epsilon$) estimated using the turbulence intensity parameter (Q; Bouffard & Boegman, 2013) indicated that η ranged from 0.05 to 0.2 in the energetic surface and bottom boundary layers, with peak mixing efficiencies in the transitional hypolimnion where $\eta = 0.2$.

3.2. Transitional Period

The transitional spring period was defined by nearly isothermal temperature profiles and weak, but stable, density stratification. Water column temperatures were actively transitioning across the maximum density threshold, with temperatures below T_{MD} at the surface ($T < T_{MD} = 3.98$°C) and above T_{MD} near the bed ($T > T_{MD} = 3.88$°C). Microstructure measurements (Figures 3 and 4) showed that average surface and bottom temperatures varied by less than 0.5°C on each day of profiling, with Brunt-Väisälä frequencies (Figure 3h) of $N^2 \approx 10^{-6}$ rad^2 / s^2. Density gradients were largely set by pressure effects, and

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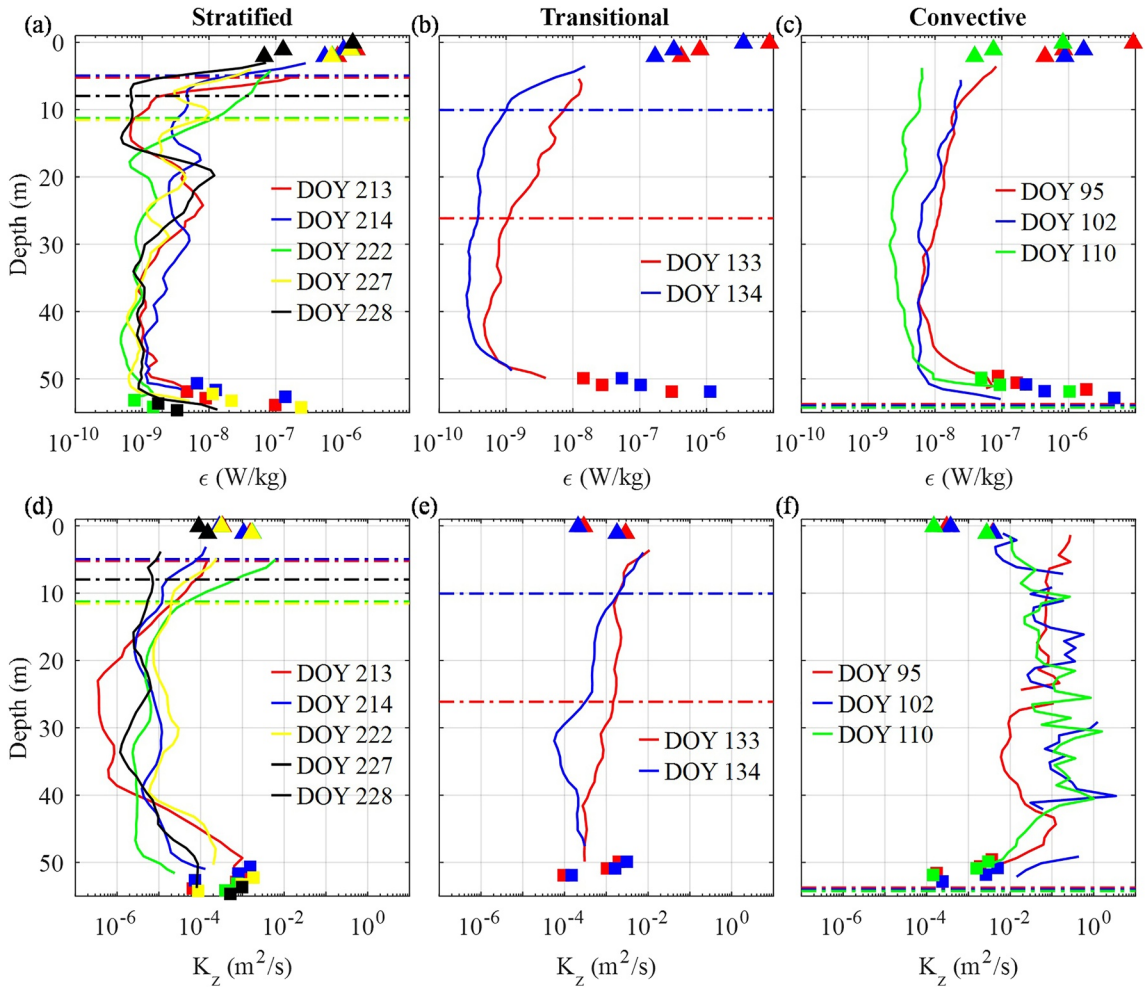


Figure 4. Average daily turbulent kinetic energy dissipation (a)–(c) and turbulent scalar diffusivity (d)–(e) profiles measured over the stratified (a, b), transitional; (b, e) and convective (c, f) periods. Individual days (day of year) are shown as different colors (stratified: 2017; transitional: 2019; convective: 2018) and SML depths are indicated with dotted-dashed lines. Surface (triangle) and bottom (square) boundary layer estimates based on law-of-the-wall scaling are included for reference (see discussion; surface: ϵ_{LOW}, $K_{z\text{-TM}}$; bottom: ϵ_{LOW}, $K_{z\text{-LOW}}$).

stratification remained stable despite the mid-depth temperature transition across T_{MD}. Sluggish currents (U = 4.7 cm / s; Figure 2e) were driven by weak winds (τ_w = 0.08 Pa; Figure 2b), and the velocity spectra (Figure 5) at 20 m depth was dominated by sub-inertial energy, with no significant contributions from near-inertial or super-inertial processes. Observed currents were less energetic than those measured over the stratified and convective periods, a result that agrees well with previous observations by Cannon and Troy (2018), who attributed this energy reduction to a decrease in storm activity and a redistribution of energy over the entire water depth. While hypolimnetic current magnitudes and mean velocity shear ($\sim 10^{-5}$ s^{-2}) were similar to those measured over the stratified period, the isothermal nature of the water column led to significant reductions in the Richardson number, with a depth-averaged mean of Ri = 0.05 and local maxima of only Ri = 0.2 in the weakly stratified surface layer. As with the stratified period, strong benthic shear led to further reductions in the Richardson number, with an average of Ri = 0.003 estimated over the bottommost 5 m.

Turbulent kinetic energy dissipation profiles measured over the transitional period (Figures 3j and 4b) were characterized by a distinct two-layer structure, with a strongly mixed surface layer (ϵ :10^{-7} –10^{-8} W / kg) separated from a more weakly energetic hypolimnion ($\epsilon \approx 8 \times 10^{-10}$ W / kg). Although wind stress estimates were similar between the stratified and transitional periods, the SML depth was significantly larger for transitional measurements (SML = 18 m), with weak surface fluxes and density stratification ($N^2 < 10^{-6}$ rad^2/s^2)

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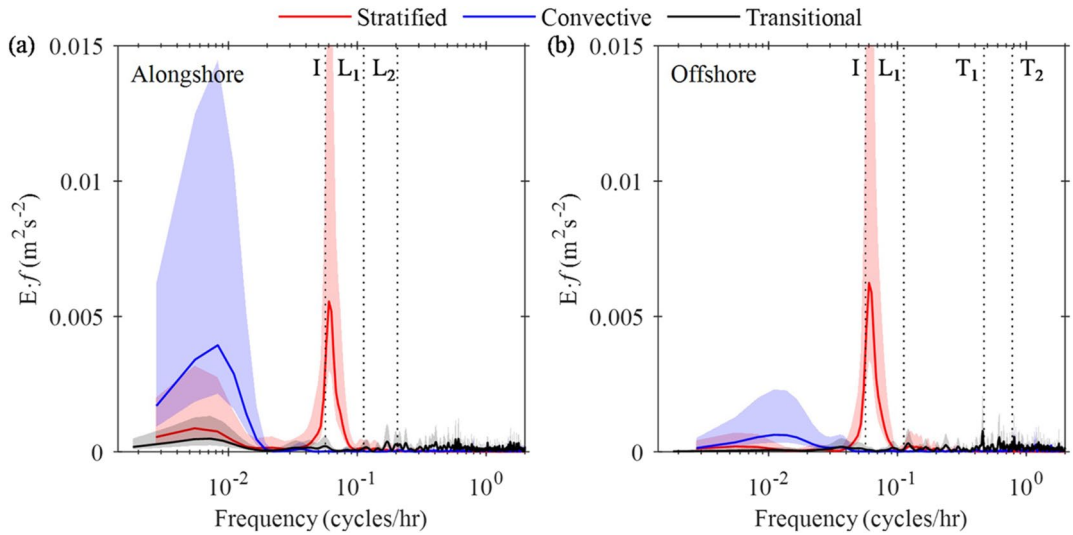


Figure 5. Variance preserving power spectrum ($E \cdot f$; m^2/s^2) for (a) alongshore and (b) offshore current velocities measured at 20 m depth. Power spectra were computed using 60 s velocity averages and a 6-day window. Potentially important frequencies are shown as vertical dashed lines following the definitions in Mortimer (2004) (I: inertial [17.5 h]; L1: longitudinal seiche [8.9 h]; L2: longitudinal seiche [4.9 h], T1: transverse seiche [2.12 h]; T2: transverse seiche [1.28 h]).

allowing wind-driven mixing to penetrate much further into the water column. While the maximum Richardson number ($Ri_{max} = 0.20$) did not exceed the typically cited critical value for stratification limited turbulence ($Ri = 1/4$), over 50% of the measurements in the surface layer fell within the "transitional regime" ($0.1 \lesssim \mathrm{Ri} \lesssim 1$; Mauritsen & Svensson, 2007), suggesting that stabilizing density stratification was able to limit the growth of the SML over this period (see discussion). Mixing rates observed during the transitional spring were significantly higher than those observed during the stratified summer (Figures 3l and 4e), with turbulent scalar diffusivities that ranged from 10^{-4} to 10^{-2} m^2/s. Although surface and bottom boundary layer mixing rates were comparable between both periods (max: $K_z \approx 10^{-2} m^2 / s$), hypolimnetic mixing was considerably more energetic in the absence of stratification, with an average diffusivity that was two orders of magnitude larger than observations made under stratified conditions (10^{-4} vs. $10^{-6} m^2 / s$). Turbulence intensity parameters (Q) were entirely within the energetic regime ($Q > 100$), and mixing efficiency (η) parameterizations (Bouffard & Boegman, 2013) indicate that η ranged from 0.05 to 0.15. Importantly, no signs of convective mixing (see next section: rapid mixed layer growth, elevated dissipation, and mixing rates) were observed during the transitional period, despite surface temperatures below T_{MD}. This suggests that minor density changes induced by daily radiative heat fluxes were insufficient to break down background stratification and drive convective mixing near the surface.

3.3. Convective Period

Observations collected over the convective period illustrate nearly isothermal waters below the temperature of maximum density with diurnal hydrodynamics that were heavily influenced by radiative convection, as previously reported by Cannon et al. (2019). Water temperatures at all depths were well below the temperature of maximum density, which varied from $T_{MD} = 3.98°C$ at the surface to $T_{MD} = 3.88°C$ at the bed (55 m depth). Persistent ice cover was not observed at the sample site at any time during the experiment. Although weakly stable ($N^2 \approx 10^{-6} rad^2 / s^2$) inverse stratification (e.g., lighter, cold water above heavier, and warm water) dominated temperature profiles in the *mean* (Figure 3n), observations showed a diurnal cycle of solar heating-induced density instabilities ($N^2 \approx -10^{-6} rad^2 / s^2$) and temperature increases during the day followed by surface cooling and re-stratification overnight (Figure 6). Horizontal velocities measured over the convective period were dominated by sub-inertial energy, which is indicative of storm-generated wind forcing on the lake. The lack of significant diurnal energy components suggests that differential heating (e.g., Bouffard & Wüest, 2019) was unimportant for driving currents during this period (Figure 5). Mean

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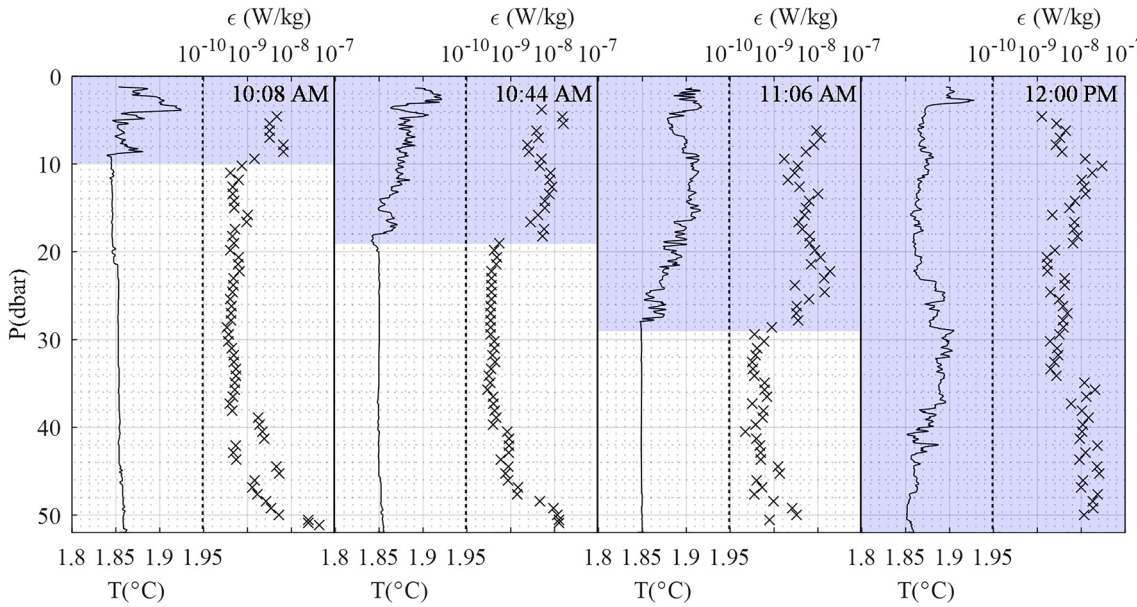


Figure 6. Representative microstructure profiles of turbulent kinetic energy dissipation ("x," W/kg) and temperature (solid line; °C) collected on April 20, 2018. Simultaneous measurements are paired sequentially, with the time of the measurement (local) included for reference. Blue boxes represent the estimated mixed layer depth (SML) and related turbulence enhancement.

velocity profiles showed very little vertical structure, with an average difference of only 1.85 ± 0.15 cm/s between velocities measured at 45 and 1 mab. Hypolimnetic Richardson number measurements followed stratification cycles, with stratification-limited estimates ($Ri_{\text{night}} > 1/4$) overnight and daily convection-driven fluctuations centered on $Ri = 0$ (positive and negative). While hypolimnetic Richardson number estimates varied dramatically throughout the day, near-bed observations showed that the *Ri* remained below ¼ for 88% of all measurements within 5 m of the bed (mean: $Ri = 0.12$), similar to observations collected during both the stratified and transitional periods.

The convective period was characterized by highly energetic, full water column mixing driven by a combination of wind shear and radiative convection. Average daily turbulent kinetic energy dissipation (Figures 3p and 4c) and turbulent scalar diffusivity (Figures 3r and 4f) profiles showed nearly constant surface-to-bed turbulence rates, with average estimates of 10^{-8} W/kg and 10^{-1} m^2/s, respectively. Modest boundary layer enhancements in observed dissipation profiles followed law-of-the-wall scaling, while the vertical structure of diffusivity near the boundaries agreed reasonably well with similarity scaling arguments. While average daily profiles of hypolimnetic dissipation showed little vertical structure, the magnitude was highly variable, scaling with both wind shear and surface fluxes. The daily SML depth encompassed the entire water column (~55 m) on each day of microstructure profiling, with full water column mixing rates (10^{-3} to 10^{0} m^2/s) that were significantly higher than those observed during the stratified or transitional period. The transitional comparison is especially important since *average* stratification ($N^2 = 6.2 \times 10^{-7}$rad^2/s^2) and wind stress ($\tau_w = 0.05$ Pa) observations were similar for measurement times during each period. These enhanced mixing rates suggest that radiative convection plays a major role in turbulence generation during the convective period, as previously reported in Cannon et al. (2019). Mixing efficiencies (η) based on stratified turbulence parameterizations (Bouffard & Boegman, 2013) indicate that average η estimates ranged from 0.01 to 0.05 during the convective period. Although the water column was stably stratified (on average) over each day of sampling, using η to describe turbulence for very weakly stratified waters under the influence of destabilizing radiative convection is questionable, and future work will involve developing mixing efficiency estimates for this forcing period.

Mixing rates varied diurnally, with convective turbulence enhancements observed during daylight hours for both diffusivity (day/night: 10^{-1} / 10^{-3} m^2/s) and dissipation (day/night: 10^{-8} / 10^{-9} W/kg). Consecutive microstructure profiles collected after sunrise (Figure 6) highlight the evolution of turbulence early in the diurnal cycle, with radiative convection eroding weak surface stratification throughout the morning, deep-

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ening the mixed layer, and driving enhanced hypolimnetic mixing across the depth by midday. Importantly, early morning hypolimnetic dissipation rates were weak ($\epsilon \approx 10^{-9}$ W / kg), with enhancements localized to the surface and bottom boundary layer until several hours after sunrise. This same cycle was also observed in turbulent scalar diffusivity, with mixing rates falling to 10^{-3} m^2/s when the water column was inversely stably stratified overnight (see Cannon et al., 2019, Figures 3 and 4). These mixing rates agree well with observations made during the transitional period, which were dynamically similar to overnight convective conditions in terms of both the density stratification ($N^2 \approx 10^{-6}$ rad^2 / s^2) and the absolute temperature gradient ($|\Delta T| \approx 0.1$°C top-to-bottom).

4. Discussion

In this study, we have described three distinct mixing periods that were observed in the deep, hypolimnetic waters of a freshwater lake. When surface temperatures were much greater than the temperature of maximum density ($T_S \gg T_{MD}$; stratified period), mixing was strongly limited by density stratification, and mixing at depth was regulated by surface stresses and hypolimnetic density gradients. When surface temperatures were below the temperature of maximum density but above freezing (convective period: 0°C $< T_S < T_{MD}$), mixing followed radiative convection cycles, with very strong mixing during the day and weak, but stable inverse stratification overnight. As the water column warmed and progressed from convective to stratified conditions ($T_S \approx T_B \approx T_{MD}$), a transitional period was observed where nearly isothermal conditions facilitated strong wind-driven mixing, but weak density gradients were still strong enough to hinder mixed layer development. Below, we discuss turbulent mixing within and below the SML, as well the implications of water depth and climate change on controlling the duration of all three mixing periods.

4.1. Surface Mixed Layer

Seasonal changes in mixing were most apparent in the highly variable surface boundary layer, where differences in density stratification and surface forcing drove dramatic changes in the magnitude and structure of turbulence profiles. These changes are best understood through an analysis of the Monin-Obukhov length (L_{MO}), which highlights the relative importance of background stratification, wind stress, and surface fluxes for generating, or limiting, near-surface turbulence during each sampling period. By definition, L_{MO} represents the depth over which wind shear dominates the production of turbulent kinetic energy (Tedford et al., 2014); at depths greater than $|L_{MO}|$, turbulence is either suppressed by stabilizing buoyancy fluxes ($L_{MO} > 0$) or enhanced by destabilizing buoyancy fluxes ($L_{MO} < 0$). Comparing observed SML depths to estimated L_{MO} allows us to evaluate which forcing mechanism (wind-driven shear and/or convection) dominates near-surface turbulence production over each observation period.

Observations of SML depths and Monin-Obukhov lengths (L_{MO}) indicate that surface mixing was largely wind-driven for all three observation periods, with destabilizing buoyancy fluxes contributing to turbulence generation during the convective winter and interior stratification limiting mixed layer growth during the transitional spring (Figure 7). For the stratified period, SML depths were small and SML $\approx L_{MO}$, suggesting that surface mixing was entirely wind-driven, with strong stabilizing surface fluxes limiting SML development to a very near-surface region. Although $|L_{MO}|$ estimates were larger than the water depth during the transitional period, the SML reached an average depth of only 17 m (SML $\ll |L_{MO}|$), highlighting the turbulence suppressing effects of weak density stratification in the interior of the water column ($Ri = 0.2$). While surface temperatures were below the temperature of maximum density during this period, radiative heating did not produce observable convective overturns. This was likely due to the negligible change in water density at temperatures near T_{MD}, where the thermal expansion coefficient (α) is minimized. In contrast, during the convective period, the deep mixed layers (SML≈depth) and generally large Monin-Obukhov lengths imply that mixing was driven by a combination of wind shear and convection. While convection did not entirely dominate mixing during any single day of profiling (SML/$L_{MO} > 10$; Tedford et al., 2014), observations of mixed layer depths greater than $|L_{MO}|$ show that convective instabilities contributed significantly to deep-water turbulence generation ($z >$ SML). Importantly, whereas weak hypolimnetic stratification limited the depth of potential wind-driven mixing during the transitional spring, this was not observed over the convective winter, despite strong similarities in overnight/early morning stratification levels. This

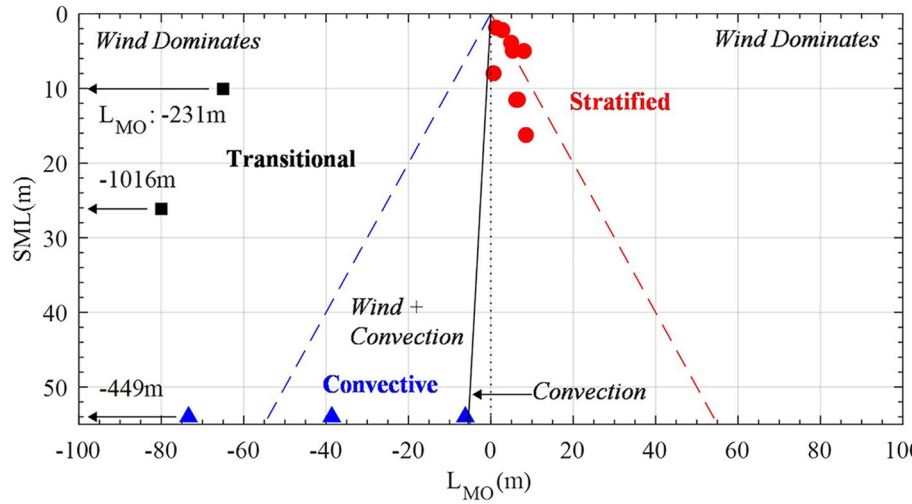


Figure 7. Average surface mixed layer depths (SML) and Monin-Obukhov lengths (L_{MO}) calculated over each day of profiling for the stratified (red circle), isothermal (black square), and convective (blue triangle) periods. 1:1 (dashed) and 1:10 (solid) lines are included for reference, with dominant turbulence generation mechanisms (wind, convection, and wind + convection) labeled following Tedford et al. (2014). L_{MO} estimates beyond axis limits are labeled and indicated with arrows.

suggests that radiative convection may be more important in its *indirect* effects on turbulence generation in the current study; namely, daily convection cycles act to quickly erode stable density stratification after sunrise, allowing wind-driven turbulence to develop uninhibited throughout the water column during the convective winter.

4.2. Below the Surface Mixed Layer

Although the vertical structure of turbulent kinetic energy dissipation varied dramatically between seasons, the mean hypolimnetic dissipation rate remained fairly constant in the absence of active convection, with an overall average of $\epsilon = 1.06 \pm 0.05 \times 10^{-9}$W/kg (± indicates 95% CI on the mean) observed outside the surface and bottom boundary layers. This average was seasonally independent, including measurements collected throughout the stratified and transitional periods, as well as before sunrise during the convective winter. This hypolimnetic dissipation rate is substantially larger than both the minimum resolvable microstructure threshold (~10^{-11} W/kg) and the ADCP-estimated laminar dissipation rate (~10^{-12} W/kg), and it likely represents shear generated background turbulence at our sample site. While velocity shear in the lake interior was persistently weak (ΔU= 1 cm/s over ~10 m), Richardson numbers were typically less 1/4 and almost always sub-unity throughout the hypolimnion. As such, the weak shear was still capable of generating mixing, albeit at rates substantially lower than those observed in the surface and bottom boundary layers. Our measurements were similar to those in other freshwater lakes, including Lake Alpnach (30 m: $\epsilon = 9 \times 10^{-9}$ W/kg; Wuest et al., 2000), Mono Lake (30 m: $\epsilon \approx 10^{-10} - 10^{-8}$ W/kg; MacIntyre et al., 1999), and Lake Erie (20 m: $\epsilon \approx 10^{-10} - 10^{-9}$ W/kg; Bouffard et al., 2012). Our observations also agree well with oceanic definitions of the background dissipation rate ($\epsilon = 10^{-9}$ W / kg: Brainerd & Gregg, 1995; Dewey & Moum, 1990; Sutherland et al., 2014), although it is significantly higher than rates measured in *very* deep freshwater lakes (Lake Baikal: Ravens et al., 2000: 10^{-11} W / kg) or the abyssal ocean (Toole et al., 1994: 10^{-11} W / kg), where velocity shear is spread out over a larger water mass. This is the first report of the background dissipation rate in Lake Michigan, providing useful context for large lake turbulence modelers.

Profiles of turbulent kinetic energy dissipation measured over the stratified period often showed elevated turbulence near the thermocline where shear and density stratification were high (Figure 4a) and observed dissipation rates were an order of magnitude larger (10^{-8} W/kg) than those measured in the weakly mixed hypolimnion (10^{-9} W/kg). This peak in dissipation was likely driven by Poincaré wave-induced thermocline shear, which is an omnipresent feature in the offshore waters of large stratified lakes (Bouffard et al., 2012; Choi et al., 2012). Figure 8 shows bulk velocity shear measured across the metalimnion

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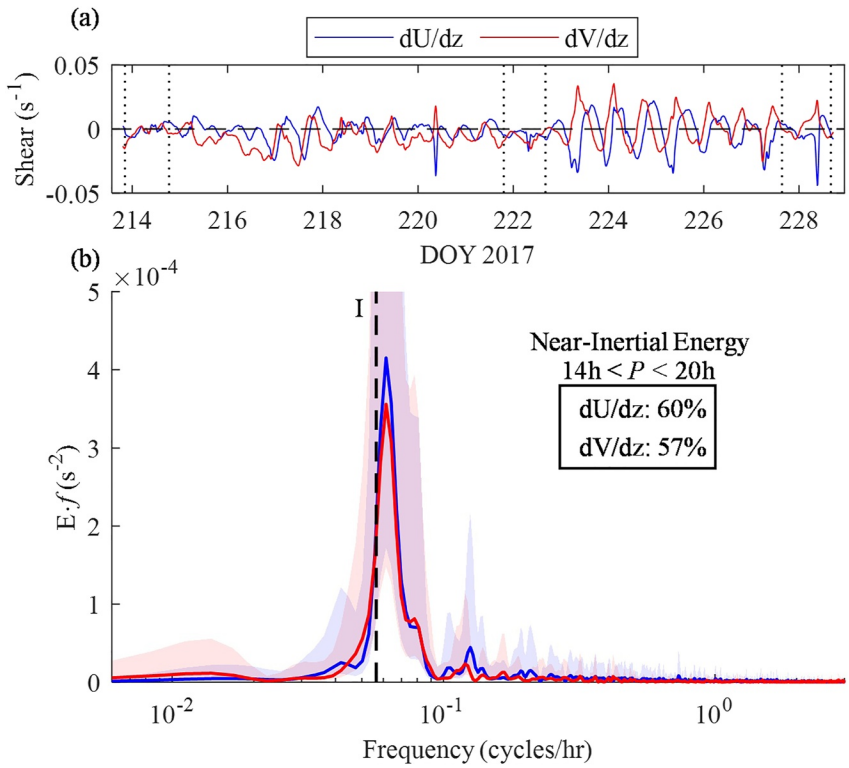


Figure 8. (a) Velocity shear time series (1 h smoothed) and (b) variance preserving power spectrum ($E \cdot f$; 1/s^2) for alongshore (dU/dz; blue) and offshore (dV/dz; red) velocity shear measured across the thermocline ($N^2 > 5 \times 10^{-4}$ rad^2 / s^2) during the stratified period. Power spectrum and 95% confidence intervals (shaded) are computed using 60 s velocity averages and a 6-day window (50% overlap). Microstructure profile times are indicated with vertical lines in (a), and the inertial period for Lake Michigan (I: $1/f = 17.5$ h) is marked with a dashed line in (b).The percent of the total variance present in the near-inertial frequency band ($14 < 1/f < 20$ h) for each shear component is included for reference.

($N^2 > 5 \times 10^{-4}$ rad^2 / s^2) during the stratified period. Thermocline shear was overwhelmingly near-inertial (Figure 8b), and nearly 60% of the total variance was contained within the near-inertial frequency band ($14 < 1/f < 20$ h). Bulk thermocline Richardson numbers (Ri_{TH}) calculated using densities and velocities from the top and bottom of the metalimnion were also sub-unity (Ri_{TH}: 0.1–0.85) suggesting that destabilizing velocity shear was present on each day of sampling (e.g., Bouffard et al., 2012). Although a full analysis of internal wave-driven instability (e.g., Lian et al., 2020) is beyond the scope of this study, these measurements provide additional evidence linking internal Poincaré wave thermocline shear to enhanced turbulent kinetic energy dissipation and turbulent mixing (Figure 4), providing an important exchange mechanism for nutrients, biota, and other constituents between the surface and bottom waters of this large, deep lake.

The vertical structure of turbulent mixing is particularly important for suspended particles, including microplastic (Zhang, 2017), phytoplankton (Huisman et al., 1999), and sediment (Y. P. Wang et al., 2013), which rely on a combination of mixing and settling for water column transport. While particles are quickly transported across the energetic surface and bottom boundary layers, where mixing rates are relatively high year-round, weak mixing in the thermocline and hypolimnion acts as a barrier to benthic particle delivery during the stratified summer (e.g., Lucas et al., 2016). This is easily illustrated with a simple time scale analysis (e.g., Koseff et al., 1993), where transport across the water depth (H) is governed by the timescales of turbulent mixing ($t_{mix} = H^2 / \overline{K_z}$) and passive settling ($t_{set} = H / \omega$) and the smaller of the two timescales represents the dominant transport timescale (t_H = minimum[t_{mix},t_{set}]). Table 2 shows delivery time scales estimated for each mixing period using representative depth-averaged turbulent diffusivities ($\overline{K_z}$) and a constant particle settling velocity ($\omega = 0.36$ m / day) consistent with passive phytoplankton sinking

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Table 2
Delivery Time Scales as Estimated for Each Mixing Period

Mixing period	Depth averaged mixing rate: $\overline{K_z}$	Time scales	
		$t_{mix} = H^2 / K_z$	$t_{set} = H / \omega$
Stratified	8.0×10^{-5} m / s	421 days	178 days
Transitional	3.6×10^{-3} m / s	9.4 days	
Convective: Day	4.9×10^{-2} m / s	17.1 h	
Convective: Night	1.4×10^{-2} m / s	2.5 days	

Note. Constant depths ($H = 55$m) and settling rates ($\omega = 3.5 \times 10^{-6}$ m/s) were used for all calculations.

in freshwater (Fahnenstiel & Scavia, 1987). Although turbulent mixing dominated vertical transport over the well-mixed transitional and convective periods (1 day $< t_H <$ 1 week), weak hypolimnetic mixing during the stratified summer ($\overline{K_z} = 8.0 \times 10^{-5}$m^2 / s) significantly increased the time required to transport particles across the water column, with transport dominated by passive settling ($t_H = 178$ days) outside the compact surface and bottom boundary layers. This annual cycle in surface-to-bed connectivity has import implications for nutrient cycling and particle transport, especially in the Laurentian Great Lakes, where the rate of benthic particle delivery directly influences the availability of phytoplankton for quagga mussel consumption, and therefore the ecosystem impact, of invasive dreissenids (e.g., Koseff et al., 1993; Vanderploeg et al., 2010).

4.3. Influence of Water Depth

The mixing periods described in this manuscript are delineated as a function of thermal stratification, with full water column turbulence characteristics that depend on both the surface temperature and hypolimnetic temperature gradients. Although, by definition, these mixing periods rely exclusively on annual stratification cycles, the relative importance of each period in the annual turbulence budget is heavily dependent on the local water depth. The seasonal transition between convective and stratified (or stratified and convective) conditions can be framed in terms of the thermal bar (e.g., Beletsky & Schwab, 2001), which forms along the boundaries of large lakes during the spring and fall turnovers. Shallow waters respond more quickly to changes in atmospheric forcing, and nearshore waters heat/cool faster than those located far from shore in the deep lake interior. This forms a temperature front (i.e., thermal bar) around the perimeter of the lake, effectively separating stratified ($T_S > T_{MD}$) and unstratified ($T_S < T_{MD}$) waters. During the spring (fall) transition, mixing conditions change from convective (stratified) to transitional to stratified (convective) following the thermal bar as it moves toward the center of the lake. This depth and therefore spatial dependence of the thermal cycle has direct consequences for the timing and duration of each mixing period throughout the lake.

The influence of water depth on the mixing period is highlighted in Figure 9, which shows surface and bottom temperature time series (2012–2013) collected at 10, 55, and 155 m depths in Lake Michigan. For simplicity, the transitional period is more loosely defined as any time when surface temperatures are greater than or approximately equal to T_{MD} and the difference between surface and bottom temperatures is less than 0.1°C. This represents forcing periods when radiative convection and stratification are both unimportant. At 55 m depth, the focus of the current study, the local stratified period lasted from May through October, with transitional mixing occurring in the late fall (November–January) and early spring (May). While the majority of the year was characterized by stratified conditions (186 days), transitional and convective mixing were still dynamically important and surface temperatures were below the temperature of maximum density for nearly 4 months (114 days). Closer to shore, the shallower (10 m depth) waters responded more quickly to atmospheric forcing, with an earlier start (mid-December) and end (mid-April) date for the convective period, consistent with cross-shore movement of the thermal bar (e.g., Beletsky & Schwab, 2001; Ullman et al., 1998). At this depth, the water column was isothermal for a large portion of the year (140 days), and mixing conditions were likely similar to those observed during the transitional period. Although the effects of ice are ignored in this manuscript, nearshore waters reached much lower temperatures than those measured offshore, and ice cover may have been hydrodynamically important for at least part of the winter (e.g., Yang et al., 2020). Far offshore in the center of the southern basin (155 m depth), surface waters were rarely mixed to the bottom, and the water column remained stratified for approximately 2/3 of the year (233 days). The strongly mixed period was short and sporadic in 2012–2013, with less than 60 total days of surface temperatures below T_{MD} and only 80 days of isothermal conditions. This lack of energetic offshore mixing is likely linked to climate change induced lake warming (discussed below), suggesting that deep convective mixing may be disappearing in the offshore waters of the Laurentian Great Lakes.

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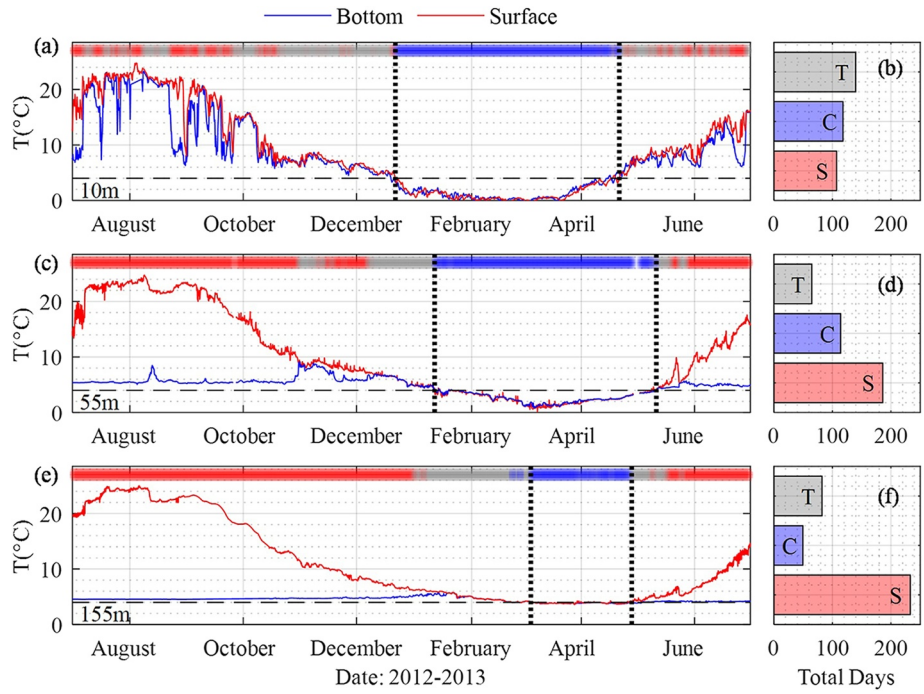


Figure 9. Time series (July 2012–2013) for surface (T_S; red line) and bottom (T_B; blue line) temperature measurements collected at 10 m (a)–(b), 55 m (c)–(d), and 155 m (e)–(f) depth. All temperature measurements were collected within 5 m of the surface and bottom boundaries, respectively. Measurements at 10 and 55 m were collected by the authors, while data at 155 m was sourced from a midlake mooring maintained by NOAA (Great Lakes Environmental Research Laboratory, 2019). Vertical lines on (a), (c), and (e) show the first and last days with surface temperatures below T_{MD}, while the total number of days experiencing potential stratified (S: $T_S > T_{MD}$; $T_S - T_B > 0.1$°C), transitional (*T*: $T_S \gtrsim T_{MD}$; $T_S - T_B < 0.1$°C), and convective (C: $T_S < T_{MD}$) mixing conditions are shown as horizontal bars (b, d, f).

4.4. The Effects of Climate Change on Offshore Mixing

Climate change has led to increasing lake surface temperatures across the globe (Woolway et al., 2020), and the world's largest lakes have proven especially vulnerable to changing environmental conditions (O'Reilly et al., 2015). In the Laurentian Great Lakes, recent studies have shown that surface and bottom temperatures are rising at an alarming rate (Anderson et al., 2021; Austin & Colman, 2007; Jabbari et al., 2021; Zhong et al., 2016), with important implications for thermal stratification and lake turnover (Fichot et al., 2019). These changes have important consequences for hypolimnetic mixing. For example, decreases in ice cover on the Great Lakes (J. Wang et al., 2012) have already drastically changed over-winter mixing, largely eliminating the relatively quiescent Winter I period (Kirillin et al., 2012) that can occur when ice at water surface shields the lake from wind and solar energy inputs. Continued increases in lake surface temperatures may also *reduce* mixing in offshore waters. In our own analysis (Figure 10) of remotely sensed lake surface temperatures (1995–2018: https://coastwatch.glerl.noaa.gov/glsea/), we found that the total annual ice-free days with surface temperatures below the temperature of maximum density have declined substantially in the deeper offshore waters of the Laurentian Great Lakes, with decreasing trends on the order of −3 days/year in the deepest waters of Lake Michigan and Lake Ontario. This annual reduction in days with surface temperatures below T_{MD} represents a substantial loss in mixing potential for offshore waters, given the disparity in mixing rates between the two periods. Although these conditions do not guarantee active, full water column mixing, especially when temperatures are very close to T_{MD} (e.g., transitional mixing) or in particularly cold or deep waters where inverse stratification may limit SML development, our observations show that surface temperatures below T_{MD} provide the highest *potential* for deep, energetic subsurface mixing. As such, the loss of convective mixing conditions may have significant effects on ecosystem functions in the Great Lakes, including primary production (Frenette et al., 1994), nutrient cycling (Goldman et al., 1996), and pelagic habitat (Kovalenko et al., 2018).

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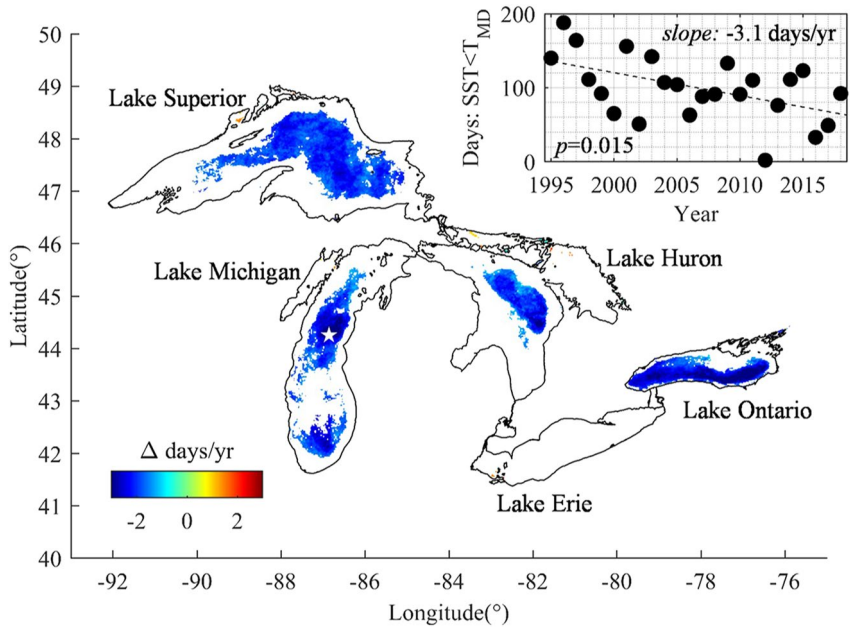


Figure 10. Changes in the total number of ice-free days with surface temperatures below the temperature of maximum density (T_{MD} = 3.98°C) in the Laurentian Great Lakes. Sea surface temperature (SST) data were sourced from Great Lakes Surface Environmental Analysis (https://coastwatch.glerl.noaa.gov/glsea/), and represent daily measurements with a spatial resolution of approximately 1.8 × 1.8 km^2. For each grid cell, the change in total days per year with temperatures below T_{MD} was estimated using a best fit line (least-squares) for data collected between 1995 and 2018 (inset diagram). Only trends with $p < 0.1$ are included in the figure.

5. Conclusion

The hypolimnetic profiles of shear, thermal stratification, turbulent kinetic energy dissipation (ϵ), and turbulent scalar diffusivity (K_z) presented in this study provide important insight into the structure and seasonal variability of hypolimnetic turbulence in large, deep lakes. Seasonal mixing characteristics varied dramatically both vertically and seasonally at the 55 m deep site, with differences linked to changes in the vertical structure of velocity shear and stratification. The structure of turbulent kinetic energy dissipation spanned three orders of magnitude ($10^{-9} - 10^{-6}$ W / kg), with seasonal variations linked to generation mechanisms unique to each forcing period, including over-winter radiative convection and (potentially) enhanced thermocline mixing generated by internal Poincarè waves during the stratified period. Hypolimnetic dissipation rates were nearly constant outside of the strongly mixed surface and bottom boundary layers, suggesting that the seasonally independent "background" dissipation rate was ~10^{-9} W/kg. While strong stratification limited both the SML depth (SML < 10 m) and hypolimnetic diffusivity ($K_z = 8 \times 10^{-7}$ m^2 / s) during the stratified summer, weak temperature gradients and wind-driven mixing drove diffusivities on the order of 10^{-3} m^2/s during the isothermal spring, with radiative convection increasing mixing rates to levels as high as 10^0 m^2/s when surface temperatures were below the temperature of maximum density. Bottom boundary stress led to elevated mixing within 5 m of the bed during all three seasons, with near-bed turbulence characteristics that agreed well with law-of-the-wall turbulence parameterizations in the mean. To the best of our knowledge, this data set is the first of its kind in the Laurentian Great Lakes, providing a valuable frame of reference for scientists concerned with hypolimnetic nutrient fluxes, biogeochemical cycles, and turbulence modeling in large, deep lakes.

Data Availability Statement

The data used in this publication are available from the Biological and Chemical Oceanography Data Management Office (BCO-DMO: Cannon et al., 2021a, 2021b, 2021c; Troy et al., 2019a, 2019b).

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